

Three segments

When pins are not on the same row or column, an L is not your only pattern

The idea

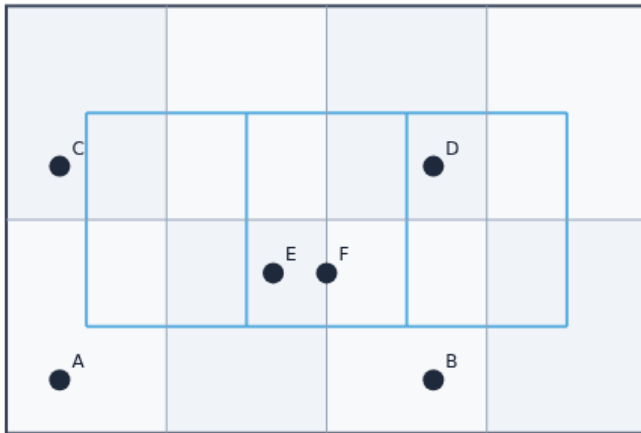
- If a and b share a row or column, fall back to L-route
- Otherwise with prefer HZ pick mid column as the average of the two columns, walk H-V-H
- Path_to_edges should show at least three edges for off-axis pairs like zero comma zero to

Two bends

Z-PATTERN ROUTE

Two bends

Step 1 / 5 · z · module02-03-pattern-z-route



Explanation

Z-route uses a midpoint column (or row) with two corners.

- vs one-bend L
- More edges possible

Toy Z

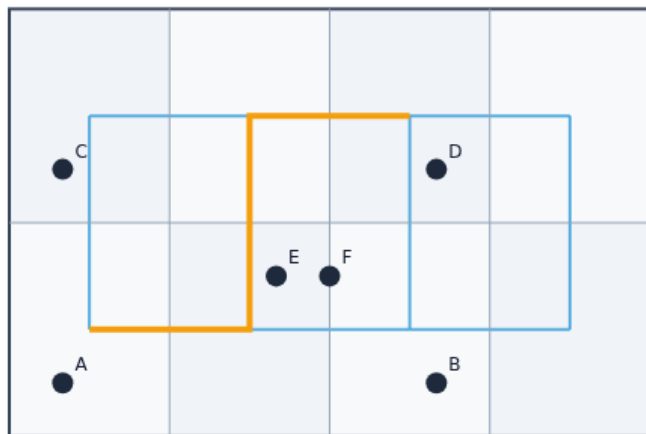
Two bends

A-D Z path

Z-PATTERN ROUTE

A-D Z path

Step 2 / 5 · ab · module02-03-pattern-z-route



Explanation

On spread, Z from A to D shows the mid-column bend.

- Compare to L
- Same pins

Highlight path

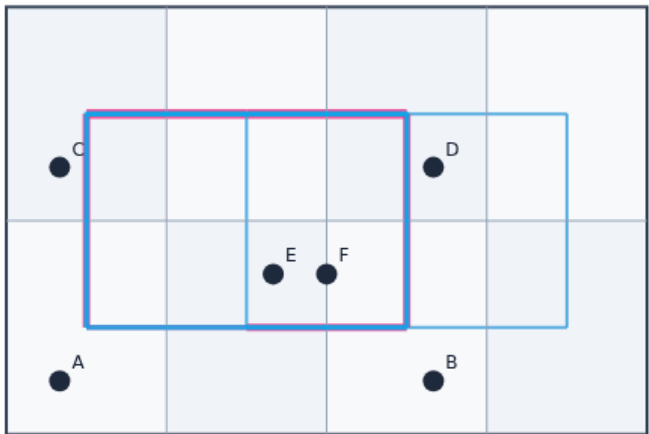
A-D Z path

L reference

Z-PATTERN ROUTE

L reference

Step 3 / 5 · 1 · module02-03-pattern-z-route



Explanation

L-HV is the fast pattern route baseline.

- Prefer HV/VH
- Pattern router

Baseline

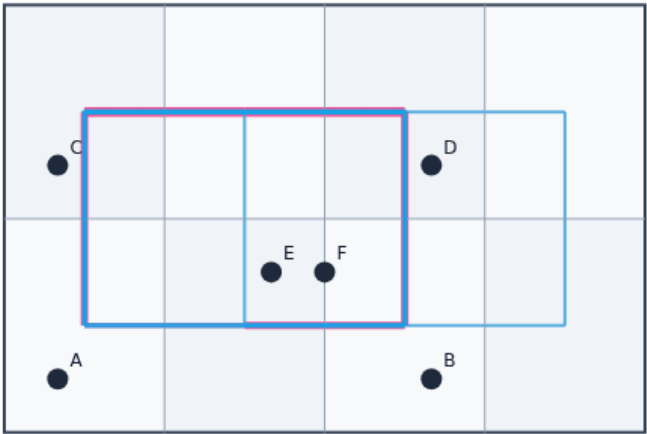
L reference

Overflow tradeoff

Z-PATTERN ROUTE

Overflow tradeoff

Step 4 / 5 · ov · module02-03-pattern-z-route



Explanation

Extra bends can spread or concentrate edge usage.

- Not always better
- Measure

Cap=2

Overflow tradeoff

Maze escape

Z-PATTERN ROUTE

Maze escape

Step 5 / 5 · next · module02-03-pattern-z-route



Explanation

When L/Z overflow, maze search detours.

- Penalize saturated edges
- Next lab

Maze

Maze escape

Browser lab track

EDA Algorithms Platform

HomeTools

HomeToolsZ vs L routing

INTERACTIVE TOOL

Z vs L routing

Compare Z-shaped two-bend routes with single-bend L routes on the same pins.

Your job: compare Z vs L on net A–B. Click *Compare Z/L*, then route.

Reset workspace

Challenges

0 / 8 cleared

Pick one

[Intro] Z uses ≥ L edges

Pick a challenge and click **Start**. Move cells, route, then **Check**. Reveal golden is study-only.

Start

Show hint

Check

Next challenge

Stop checking

Idle

A

B

C

D

E

F

←

→

↑

↓

Reset to starter

Reveal golden (study)

Route L–HV

Route L–VH

Route maze

Rip-up reroute

Compare Z/L on A–B

Graph

Reference / metrics

selected: A @ GCell (0,0)

routes: (none – click Route)

overflow total: 0.00 · max: 0.00 · count: 0

edge capacity: 2

HPWL: 34.0

view: your placement

Browser lab starter

learn_global_routing — 03 pattern z route

Implement track

- Implement `z_route(a, b, prefer)``
- Print Z and L paths for the same terminal pair; explain one edge difference

Pitfalls

- Using a midpoint that leaves zero-length segments
- Not falling back to L on aligned pins
- Confusing Z prefer HZ with L prefer HV naming

Your turn

- Finish Z routing on two-pin nets
- Next: maze routing that respects congested edges